FORMULE

v 1.3

FAZORI

$$u(t) = U_{max}\, cos(\omega t + \theta) \qquad \underline{U} = \frac{U_{max}}{\sqrt{2}} e^{j\theta} \qquad u(t) = \sqrt{2}\, Re\{\underline{U} e^{j\omega t}\}$$

EFEKTIVNA VRIJEDNOST

$$I^{eff} = \frac{1}{T}\sqrt{\int_0^T i^2(t)dt}$$

SNAGA

$$p(t) = UI\, cos\,\varphi + UI\, cos(2\omega t + \theta + \beta) \qquad \underline{S} = \underline{U}\,\underline{I}^*$$

FORTESCUEOVA TRANSFORMACIJA

$$\begin{bmatrix} \underline{I}_a \\ \underline{I}_b \\ \underline{I}_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ \underline{a}^2 & \underline{a} & 1 \\ \underline{a} & \underline{a}^2 & 1 \end{bmatrix} \cdot \begin{bmatrix} \underline{I}_d \\ \underline{I}_i \\ \underline{I}_0 \end{bmatrix} \qquad \begin{bmatrix} \underline{I}_d \\ \underline{I}_i \\ \underline{I}_0 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & \underline{a} & \underline{a}^2 \\ 1 & \underline{a}^2 & \underline{a} \\ 1 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} \underline{I}_a \\ \underline{I}_b \\ \underline{I}_c \end{bmatrix} \qquad \underline{a} = e^{j\frac{2\pi}{3}} = -\frac{1}{2} + j\frac{\sqrt{3}}{2}$$

TRANSFORMACIJE ZVIJEZDA - TROUGAO

$$\underline{Z}_1 = \frac{\underline{Z}_{12}\underline{Z}_{13}}{\underline{Z}_{12} + \underline{Z}_{13} + \underline{Z}_{23}} \qquad \underline{Z}_2 = \frac{\underline{Z}_{12}\underline{Z}_{23}}{\underline{Z}_{12} + \underline{Z}_{13} + \underline{Z}_{23}} \qquad \underline{Z}_3 = \frac{\underline{Z}_{13}\underline{Z}_{23}}{\underline{Z}_{12} + \underline{Z}_{13} + \underline{Z}_{23}}$$

$$\underline{Z}_{12} = \underline{Z}_1 + \underline{Z}_2 + \frac{\underline{Z}_1\underline{Z}_2}{\underline{Z}_3} \qquad \underline{Z}_{13} = \underline{Z}_1 + \underline{Z}_3 + \frac{\underline{Z}_1\underline{Z}_3}{\underline{Z}_2} \qquad \underline{Z}_{23} = \underline{Z}_2 + \underline{Z}_3 + \frac{\underline{Z}_2\underline{Z}_3}{\underline{Z}_1}$$

PAD NAPONA NA REDNOJ GRANI

$$\underline{U}_1 = U_2 + \frac{P_2R + Q_2X}{U_2} + j\frac{P_2X - Q_2R}{U_2} = U_2 + \Delta U_2 + j\delta U_2$$

$$P^{gub} = R\frac{P^2 + Q^2}{U^2} \qquad Q^{gub} = X\frac{P^2 + Q^2}{U^2}$$

MODEL VODA

$$D_{SG} = \sqrt[3]{D_{ab} D_{bc} D_{ca}} \qquad r_{es} = \sqrt[n]{n r_e R^{n-1}} \qquad r_V = \frac{\rho}{s}$$

$$l_V = 2 \cdot 10^{-4} \, ln\left(\frac{D_{SG}}{r_{es}}\right) \qquad c_V = \frac{55{,}55 \cdot 10^{-9}}{ln\left(\frac{D_{SG}}{r_{es}^c} \cdot \frac{2h_{SG}}{\sqrt{4h_{SG}^2 + D_{SG}^2}}\right)} \qquad g_V = 10^{-4}\left[1 - e^{-3{,}05\left(U_{f,r}/U_{f,kr} - 1\right)}\right]$$

$$\underline{\gamma} = \sqrt{\underline{z}_V \, \underline{y}_V} = \alpha + j\beta \qquad \underline{Z}_C = \sqrt{\frac{\underline{z}_V}{\underline{y}_V}} \qquad P_{nat} = \frac{U_n^2}{Z_C}$$

$$\underline{U}(x) = ch(\underline{\gamma} x)\underline{U}_2 + Z_C sh(\underline{\gamma} x)\underline{I}_2 \qquad \underline{I}(x) = \frac{1}{\underline{Z}_C} sh(\underline{\gamma} x)\underline{U}_2 + ch(\underline{\gamma} x)\underline{I}_2$$

$$k_R = 1 - \frac{X_V B_V}{3} \qquad k_X = 1 - \frac{1}{6} X_V B_V \left(1 - \frac{R_V^2}{X_V^2}\right) \qquad k_B = 1 + \frac{X_V B_V}{12}$$

$$\underline{k}_Z = \frac{sh\sqrt{\underline{Z}_V \underline{Y}_V}}{\sqrt{\underline{Z}_V \underline{Y}_V}} \qquad \underline{k}_Y = \frac{2\left(ch\sqrt{\underline{Z}_V \underline{Y}_V} - 1\right)}{\sqrt{\underline{Z}_V \underline{Y}_V} \, sh\sqrt{\underline{Z}_V \underline{Y}_V}}$$

MODEL TRANSFORMATORA

$$N_1 i_1 - N_2 i_2 = R_m \Psi \qquad m_n = \frac{U_{n1}}{U_{n2}} \qquad Z_T = \frac{u_k[\%]}{100} \frac{U_{n1}^2}{S_n} \qquad Z_T^{(2)} = \frac{u_k[\%]}{100} \frac{U_{n2}^2}{S_n}$$

$$R_T = P_{Cun}^{gub}\left(\frac{U_{n1}}{S_n}\right)^2 \qquad R_T^{(2)} = P_{Cun}^{gub}\left(\frac{U_{n2}}{S_n}\right)^2 \qquad X_T = \sqrt{Z_T^2 - R_T^2} \qquad X_T^{(2)} = \sqrt{Z_T^{(2)2} - R_T^{(2)2}}$$

$$Y_{PH} = \frac{i[\%]}{100} \frac{S_n}{U_{n1}^2} \qquad Y_{PH}^{(2)} = \frac{i[\%]}{100} \frac{S_n}{U_{n2}^2} \qquad G = \frac{P_{Fen}^{gub}}{U_{n1}^2} \qquad G^{(2)} = \frac{P_{Fen}^{gub}}{U_{n2}^2}$$

$$B = \sqrt{Y_{PH}^2 - G^2} \qquad B^{(2)} = \sqrt{Y_{PH}^{(2)2} - G^{(2)2}}$$

$$Z_{1,2} = \frac{u_{k1,2}[\%]}{100} \cdot \frac{U_{n1}^2}{S_{1,2}} \qquad Z_{1,3} = \frac{u_{k1,3}[\%]}{100} \cdot \frac{U_{n1}^2}{S_{1,3}} \qquad Z_{2,3} = \frac{u_{k2,3}[\%]}{100} \cdot \frac{U_{n1}^2}{S_{2,3}}$$

$$\underline{Z}_{1,2} = \underline{Z}_1 + \underline{Z}_2^{(1)} \qquad \underline{Z}_{1,3} = \underline{Z}_1 + \underline{Z}_3^{(1)} \qquad \underline{Z}_{2,3} = \underline{Z}_2^{(1)} + \underline{Z}_3^{(1)}$$

$$\underline{Z}_1 = \frac{1}{2}\left(\underline{Z}_{1,2} + \underline{Z}_{1,3} - \underline{Z}_{2,3}\right) \qquad \underline{Z}_2^{(1)} = \frac{1}{2}\left(\underline{Z}_{1,2} - \underline{Z}_{1,3} + \underline{Z}_{2,3}\right) \qquad \underline{Z}_3^{(1)} = \frac{1}{2}\left(-\underline{Z}_{1,2} + \underline{Z}_{1,3} + \underline{Z}_{2,3}\right)$$

MODEL REGULACIONOG TRANSFORMATORA

$$m = m_n t = m_n (1 + n\Delta t)$$

$$\underline{Z}_1 = \frac{t^2}{1-t}\underline{Z}_T \qquad \underline{Z}_2 = t\underline{Z}_T \qquad \underline{Z}_3 = \frac{t}{t-1}\underline{Z}_T$$

MODEL POTROŠNJE

$$P_P = C_{PU} U^{k_{PU}} \qquad Q_P = C_{QU} U^{k_{QU}}$$

$$k_{PU\sum} = k_{PU_I}\frac{P_{P_I}}{P_P} + k_{PU_{II}}\frac{P_{P_{II}}}{P_P} + k_{PU_{III}}\frac{P_{P_{III}}}{P_P} + \cdots \qquad k_{QU\sum} = k_{QU_I}\frac{Q_{P_I}}{Q_P} + k_{QU_{II}}\frac{Q_{P_{II}}}{Q_P} + k_{QU_{III}}\frac{Q_{P_{III}}}{Q_P} + \cdots$$

DIJAGRAM POTROŠNJE

$$W_d = \int_0^{24h} P(t)dt = P_{sr} \cdot 24h \qquad m = \frac{W_d}{24P_{max}} = \frac{P_{sr}}{P_{max}} \qquad T_e = \frac{W_d}{P_{max}} \qquad T_e^{god} = \frac{W_{god}}{P_{max}^{god}}$$

RAST POTROŠNJE NA VREMENSKOM HORIZONTU

$$W_n = W_o (1+p)^n \qquad W = W_o e^{at} \qquad P = P_o e^{bt}$$

METOD „KORAK PO KORAK” ZA GRAĐENJE [$\underline{Z}_{BUS}$]

$$\underline{Z}^{post} = \left[\begin{array}{ccc:c} & & & 0 \\ & \underline{Z}^{pre} & & 0 \\ & & & \vdots \\ \hdashline 0 & 0 & \cdots & \underline{Z}_{gr} \end{array}\right] \qquad \underline{Z}^{post} = \left[\begin{array}{ccc:c} & & & \underline{Z}_{1j} \\ & \underline{Z}^{pre} & & \underline{Z}_{2j} \\ & & & \vdots \\ \hdashline \underline{Z}_{j1} & \underline{Z}_{j2} & \cdots & \left(\underline{Z}_{gr} + \underline{Z}_{jj}\right) \end{array}\right]$$

$$\underline{Z}^{post} = \underline{Z}^{pre} - \frac{1}{\underline{Z}_{jj} + \underline{Z}_{gr}} \begin{bmatrix} \underline{Z}_{1j} \\ \underline{Z}_{2j} \\ \vdots \\ \underline{Z}_{nj} \end{bmatrix} \times \begin{bmatrix} \underline{Z}_{j1} & \underline{Z}_{j2} & \cdots & \underline{Z}_{jn} \end{bmatrix}$$

$$\underline{Z}^{post} = \underline{Z}^{pre} - \frac{1}{\underline{Z}_{gr} + \underline{Z}_{ii} + \underline{Z}_{jj} - 2\underline{Z}_{ij}} \begin{bmatrix} \underline{Z}_{1i} - \underline{Z}_{1j} \\ \underline{Z}_{2i} - \underline{Z}_{2j} \\ \vdots \\ \underline{Z}_{ni} - \underline{Z}_{nj} \end{bmatrix} \times \begin{bmatrix} \underline{Z}_{i1} - \underline{Z}_{j1} & \underline{Z}_{i2} - \underline{Z}_{j2} & \cdots & \underline{Z}_{in} - \underline{Z}_{jn} \end{bmatrix}$$

MATRIČNA METODA ZA PRORAČUN KRATKIH SPOJEVA

$$\underline{I}_{qk} = \frac{\underline{U}_{qr}}{\underline{Z}_k + \underline{Z}_{qq}} \qquad \underline{U}_{ik} = \underline{U}_{ir} - \frac{\underline{Z}_{iq}}{\underline{Z}_k + \underline{Z}_{qq}} \underline{U}_{qr} \qquad i = 1,2,\ldots,n \quad (i \neq q)$$

UTICAJ KRATKIH SPOJEVA NA TELEKOMUNIKACIONE VODOVE

$$E[V] = j\omega \cdot \left(3I_o[kA]\right) \cdot \frac{4}{\sqrt{a[m]}} \cdot l_{\Pi}[km] \cdot k_r$$

GAUSS-SEIDELOV METOD

$$\underline{U}_i^{(\nu+1)} = \frac{1}{\underline{Y}_{ii}}\left[\left(\frac{\underline{S}_i}{\underline{U}_i^{(\nu)}}\right)^* - \sum_{k=1}^{i-1}\underline{U}_k^{(\nu+1)}\cdot\underline{Y}_{ik} - \sum_{k=i+1}^{n}\underline{U}_k^{(\nu)}\cdot\underline{Y}_{ik}\right]; \qquad i=\overline{1:n}$$

$$Q_i^{(\nu)} = -Jm\left\{\underline{Y}_{ii}U_i^2 + \sum_{\substack{j=1\\ j\neq i}}^{i-n}\underline{Y}_{ij}\cdot\underline{U}_j^{(\nu)}\cdot\underline{U}_i^{'(\nu)*}\right\}; \qquad i=\overline{1:\mathrm{m}}$$

$\underline{U}_i^{'}$ – napon čiji je poznat (specificiran) fazni stav.

NEWTON-RAPHSONOV METOD

$$\left.\begin{aligned} p_i &= \sum_{k=1}^{n}u_i\cdot u_k\cdot\left[g_{ik}\cdot\cos\left(\theta_i-\theta_k\right)+b_{ik}\cdot\sin\left(\theta_i-\theta_k\right)\right]\\ q_i &= \sum_{k=1}^{n}u_i\cdot u_k\cdot\left[g_{ik}\cdot\sin\left(\theta_i-\theta_k\right)-b_{ik}\cdot\cos\left(\theta_i-\theta_k\right)\right]\end{aligned}\right\} \qquad i=\overline{1,n}$$

$$\left.\begin{aligned} \Delta p_i &= p_i^{spec}-p_i = p_i^{spec}-\sum_{k=1}^{n}u_i\cdot u_k\cdot\left[g_{ik}\cdot\cos\left(\theta_i-\theta_k\right)+b_{ik}\cdot\sin\left(\theta_i-\theta_k\right)\right]\\ \Delta q_i &= q_i^{spec}-q_i = q_i^{spec}-\sum_{k=1}^{n}u_i\cdot u_k\cdot\left[g_{ik}\cdot\sin\left(\theta_i-\theta_k\right)-b_{ik}\cdot\cos\left(\theta_i-\theta_k\right)\right]\end{aligned}\right\} \qquad i=\overline{1,n}$$

$$\frac{\partial\Delta p_i}{\partial\theta_i} = \sum_{\substack{k=1\\ k\neq i}}^{n}u_i\cdot u_k\cdot\left[g_{ik}\cdot\sin\left(\theta_i-\theta_k\right)-b_{ik}\cdot\cos\left(\theta_i-\theta_k\right)\right]$$

$$\frac{\partial\Delta p_i}{\partial\theta_j} = u_i\cdot u_j\cdot\left[-g_{ij}\cdot\sin\left(\theta_i-\theta_j\right)+b_{ij}\cdot\cos\left(\theta_i-\theta_j\right)\right], j\neq i$$

$$\frac{\partial\Delta p_i}{\partial u_i} = -\sum_{\substack{k=1\\ k\neq i}}^{n}u_k\cdot\left[g_{ik}\cdot\cos\left(\theta_i-\theta_k\right)+b_{ik}\cdot\sin\left(\theta_i-\theta_k\right)\right]-2\cdot u_i\cdot g_{ii}$$

$$\frac{\partial\Delta p_i}{\partial u_j} = -u_i\cdot\left[g_{ij}\cdot\cos\left(\theta_i-\theta_j\right)+b_{ij}\cdot\sin\left(\theta_i-\theta_j\right)\right], j\neq i$$

$$\frac{\partial\Delta q_i}{\partial\theta_i} = -\sum_{\substack{k=1\\ k\neq i}}^{n}u_i\cdot u_k\cdot\left[g_{ik}\cdot\cos\left(\theta_i-\theta_k\right)+b_{ik}\cdot\sin\left(\theta_i-\theta_k\right)\right]$$

$$\frac{\partial\Delta q_i}{\partial\theta_j} = u_i\cdot u_j\cdot\left[g_{ij}\cdot\cos\left(\theta_i-\theta_j\right)+b_{ij}\cdot\sin\left(\theta_i-\theta_j\right)\right], j\neq i$$

$$\frac{\partial\Delta q_i}{\partial u_i} = -\sum_{\substack{k=1\\ k\neq i}}^{n}u_k\cdot\left[g_{ik}\cdot\sin\left(\theta_i-\theta_k\right)-b_{ik}\cdot\cos\left(\theta_i-\theta_k\right)\right]+2\cdot u_i\cdot b_{ii}$$

$$\frac{\partial\Delta q_i}{\partial u_j} = -u_i\cdot\left[g_{ij}\cdot\sin\left(\theta_i-\theta_j\right)-b_{ij}\cdot\cos\left(\theta_i-\theta_j\right)\right], j\neq i$$

BRZI STOTTOV METOD

$$\frac{\partial \Delta p_i}{\partial \theta_i} = -u_i \cdot \sum_{\substack{k=1 \\ k \neq i}}^{n} b_{ik}$$

$$\frac{\partial \Delta p_i}{\partial \theta_j} = u_i \cdot b_{ij} \quad , j \neq i$$

$$\frac{\partial \Delta q_i}{\partial u_i} = u_i \cdot \left(2 \cdot b_{ii} + \sum_{\substack{k=1 \\ k \neq i}}^{n} b_{ik} \right)$$

$$\frac{\partial \Delta q_i}{\partial u_j} = u_i \cdot b_{ij} \quad , j \neq i$$

$$b'_{ij} = \begin{cases} -\sum_{\substack{k=1 \\ k \neq i}}^{n} b_{ik}, & i = j \\ b_{ij}, & i \neq j \end{cases}$$

$$b''_{ij} = \begin{cases} 2 \cdot b_{ii} + \sum_{\substack{k=1 \\ k \neq i}}^{n} b_{ij}, & i = j \\ b_{ij}, & i \neq j \end{cases}$$

DC METOD

$$p_{ik} = \frac{\theta_i - \theta_k}{x_{ik}^{gr}}$$

TOKOVI SNAGA

$$p_{ik} = u_i^2 \cdot \left(g_i^0 + g_{ik}^{gr} \right) - u_i \cdot u_k \cdot \left[g_{ik}^{gr} \cdot \cos\left(\theta_i - \theta_k \right) + b_{ik}^{gr} \cdot \sin\left(\theta_i - \theta_k \right) \right]$$

$$q_{ik} = -u_i^2 \cdot \left(b_i^0 + b_{ik}^{gr} \right) - u_i \cdot u_k \cdot \left[g_{ik}^{gr} \cdot \sin\left(\theta_i - \theta_k \right) - b_{ik}^{gr} \cdot \cos\left(\theta_i - \theta_k \right) \right]$$

$$p_{gub} = p_{ik} + p_{ki} \qquad \sum p_{gub} = \sum \left(p_{Gi} - p_{Pi} \right)$$

$$q_{gub} = q_{ik} + q_{ki} \qquad \sum q_{gub} = \sum \left(q_{Gi} - q_{Pi} \right) + \sum u_i^2 \cdot b_{ii}$$